

Assignment 5: LoRA Fine-Tuning and Adversarial Robustness

Ravi Sharma
M25CSA024

1 Introduction

In this assignment, we explore parameter-efficient fine-tuning using LoRA on a Vision Transformer (ViT-S) model for CIFAR-100 classification. Additionally, adversarial robustness is analyzed using IBM ART.

2 Q1: LoRA Fine-Tuning

2.1 Baseline Model

A pre-trained ViT-S model was fine-tuned on CIFAR-100 without LoRA.

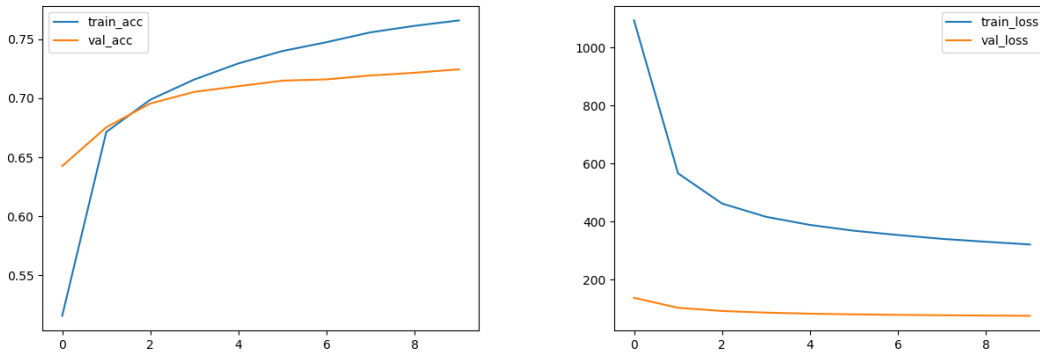


Figure 1: Baseline Training Curves

2.2 LoRA Experiments

LoRA was applied to attention layers (Q, K, V) with:

- Rank: 2, 4, 8
- Alpha: 2, 4, 8
- Dropout: 0.1

Experiments were conducted across all combinations.

LoRA	Rank	Alpha	Dropout	Test Acc	Trainable Params
Without LoRA	-	-	-	0.72	38,500
With LoRA	2	2	0.1	0.65	259,684
With LoRA	2	4	0.1	0.63	259,684
With LoRA	4	4	0.1	0.73	259,684
With LoRA	4	8	0.1	0.73	259,684
With LoRA	8	8	0.1	0.84	259,684

Table 1: Comparison of LoRA Configurations

2.3 Results Table

2.4 Best Configuration

The best performance was achieved with:

- Rank = 8
- Alpha = 8

2.5 Training Curves (Best Model)

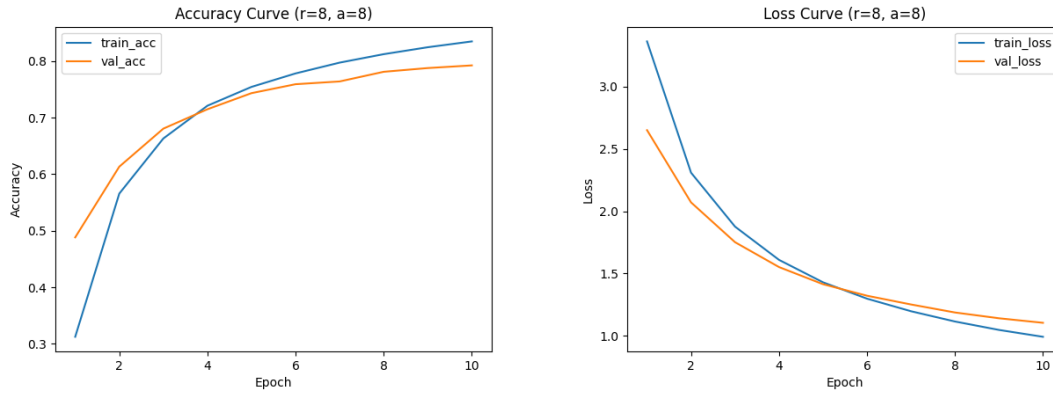


Figure 2: Training and Validation Curves (r=8, a=8)

2.6 Gradient Analysis

2.7 Class-wise Accuracy

2.8 Observations

- Increasing rank and alpha improves model capacity.
- LoRA significantly reduces trainable parameters while maintaining strong performance.
- The best configuration achieves strong generalization with minimal overfitting.

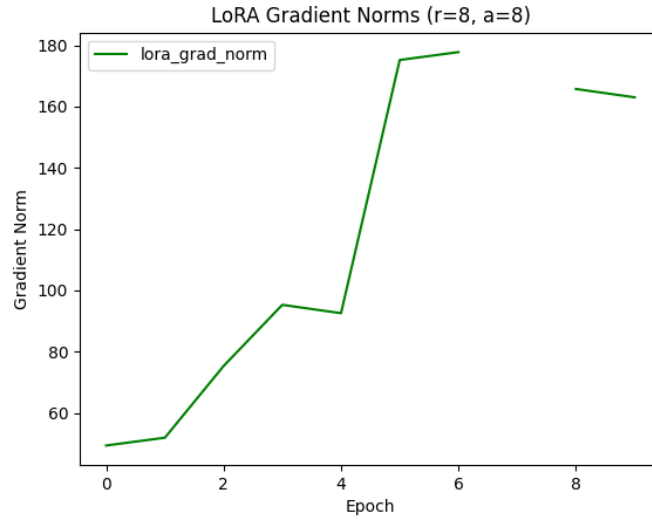


Figure 3: LoRA Gradient Norms

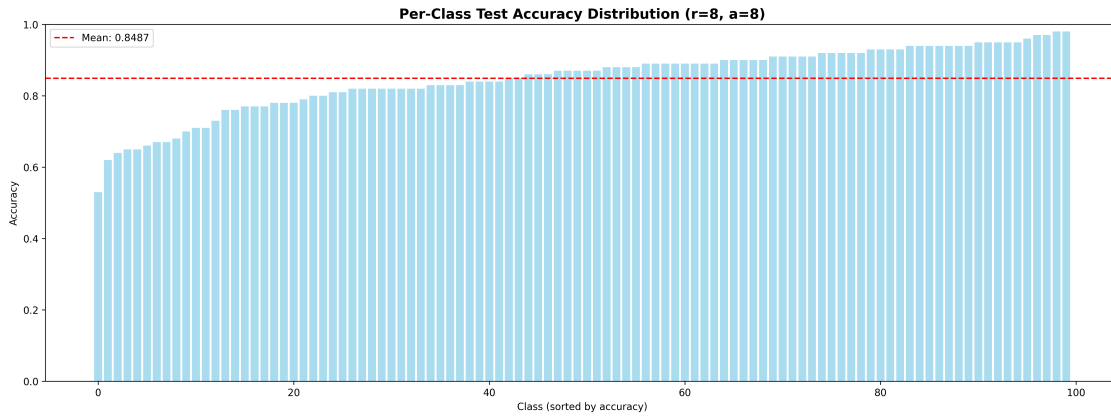


Figure 4: Per-Class Accuracy Distribution

2.9 Optuna Hyperparameter Search

Optuna was used to automate hyperparameter tuning over rank and alpha. The optimal configuration found was $(r=8, a=8)$, consistent with manual experimentation.

3 Links

- GitHub Repository: [GitHub Repository](#)
- WandB Dashboard: [WandB Dashboard](#)
- Huggingface model: [Huggingface model](#)

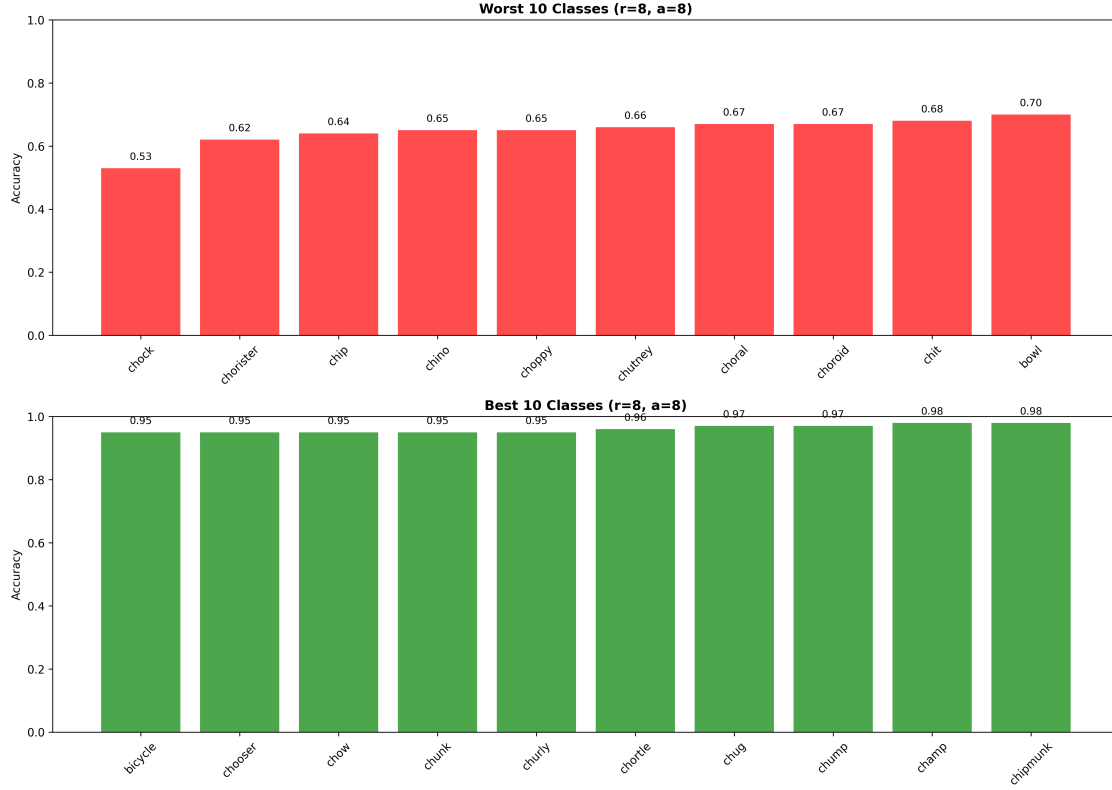


Figure 5: Best and Worst Performing Classes

4 Conclusion

LoRA enables efficient fine-tuning with significantly fewer trainable parameters while achieving competitive performance. The approach scales well with increasing rank and alpha, with (8,8) providing the best trade-off.

5 Q2: Adversarial Attacks and Detection using IBM ART

5.1 FGSM Attack: From Scratch vs IBM ART

5.1.1 Objective

The objective is to evaluate the robustness of a ResNet18 model trained on CIFAR-10 against adversarial attacks and compare FGSM implementations with and without IBM ART.

5.1.2 Model Performance

The ResNet18 model achieved a clean test accuracy of **82.64%**.

5.1.3 FGSM Results

Epsilon	FGSM (Scratch)	FGSM (ART)
0.00	0.8264	0.8264
0.01	0.5510	0.55
0.03	0.1873	0.18
0.05	0.0655	0.06
0.10	0.0120	0.01

Table 2: FGSM Attack Performance

5.1.4 FGSM Plots

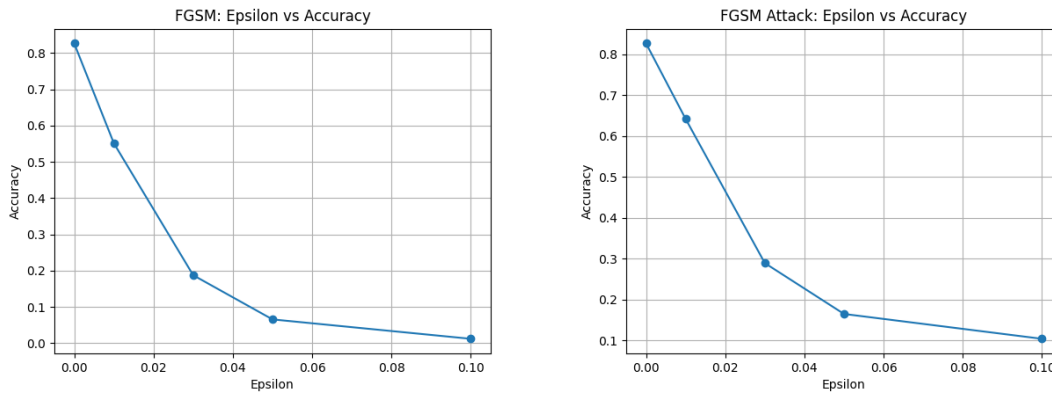


Figure 6: FGSM: Epsilon vs Accuracy (Scratch vs ART)

5.1.5 Observations

- Accuracy decreases sharply with increasing perturbation strength.
- FGSM implementations (scratch and ART) produce consistent results.
- Even small perturbations significantly degrade model performance.

5.2 Adversarial Detection Model

5.2.1 Objective

To design a ResNet34-based detector that distinguishes between clean and adversarial images.

5.2.2 PGD Detection

- Attack: Projected Gradient Descent (IBM ART)
- Detection Accuracy: **98.15%**

5.2.3 BIM Detection

- Attack: Basic Iterative Method (IBM ART)
- Detection Accuracy: **97.42%**

5.2.4 Detection Comparison

Attack	Detection Accuracy (%)
PGD (IBM ART)	98.15
BIM (IBM ART)	97.42

Table 3: Detection Performance Comparison

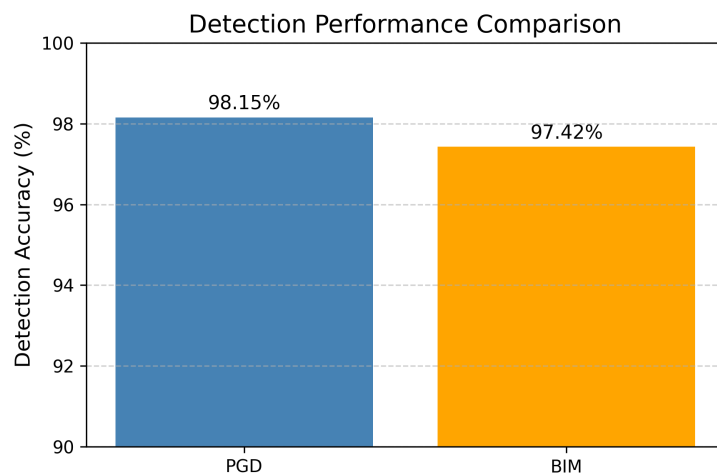


Figure 7: Comparison of Detection Accuracy

5.2.5 Observations

- Detection accuracy exceeds 97% for both PGD and BIM attacks.
- PGD perturbations are slightly easier to detect than BIM.

- The detector generalizes well across different adversarial attack types.

5.3 WandB Visualization

To qualitatively analyze adversarial robustness, 10 samples were logged on WandB showing:

- Clean images
- FGSM (Scratch)
- FGSM (IBM ART)
- PGD (IBM ART)
- BIM (IBM ART)

WandB Dashboard: [WandB Link](#)

The image shows a screenshot of a WandB dashboard table. The table has 5 columns: 'Clean', 'FGSM_Scratch', 'FGSM_ART', 'PGD_ART', and 'BIM_ART'. There are 10 rows, each representing a sample. Each row contains a small thumbnail image for each of the five categories. The 'Clean' column shows original images of various objects like a boat, a landscape, a person, and a car. The other columns show the corresponding images after applying different adversarial attack methods. The interface includes a 'Filter' button at the top left, a pagination bar at the bottom showing '1 - 10 of 10', and links for 'Export as CSV', 'Columns...', and 'Reset table'.

	Clean	FGSM_Scratch	FGSM_ART	PGD_ART	BIM_ART
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					

Figure 8: 10 samples were logged on WandB

5.4 Conclusion (Q2)

- Adversarial attacks significantly reduce model accuracy.
- FGSM implementations from scratch and ART are consistent.
- Detection models effectively identify adversarial samples with high accuracy.
- The system is robust across multiple attack types.